CS471 — Labs 3, 4 & 5

Django URLs & Templates · HTML/CSS Fundamentals · Static Integration

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Course:	CS471
GitHub Repository:	https://github.com/Thana111/CS471-Labs

Lab 3 — Django URLs, Views & Templates

This lab covers mapping URLs to view functions, passing parameters via GET requests and URL paths, creating a basic HTML template, rendering context variables, splitting URL configuration per app, and building a dynamic book-detail view.

Task 1 & 2 — First view, URL mapping, and GET parameters

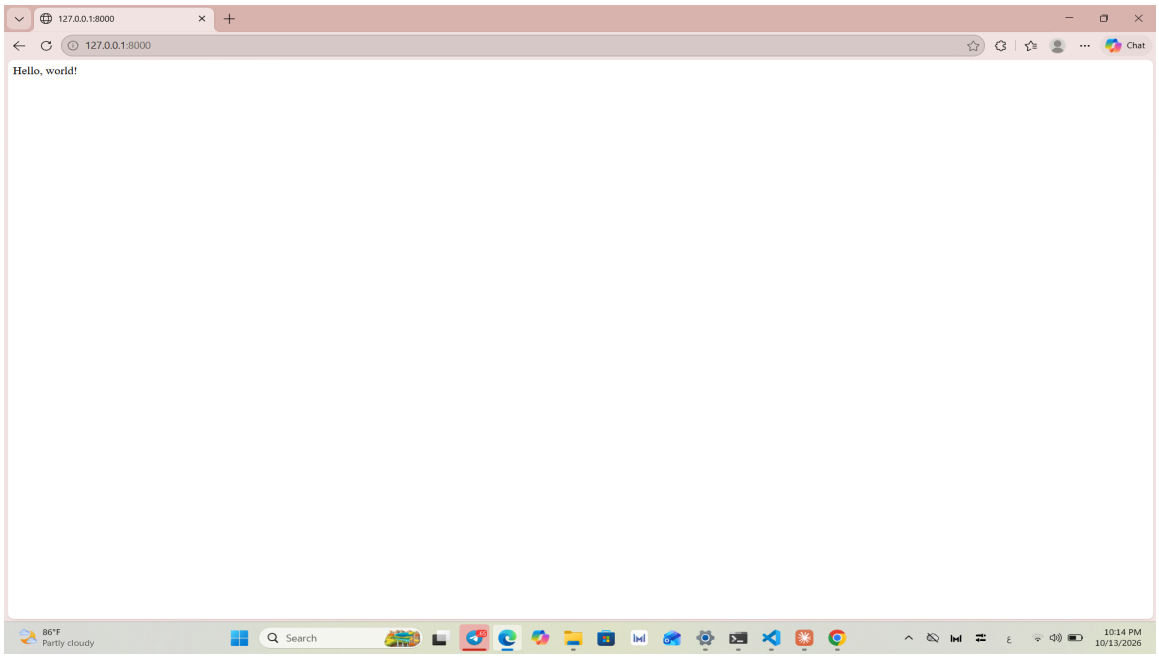


Figure 1: http://127.0.0.1:8000 → 'Hello, world!' (default view, no parameter)

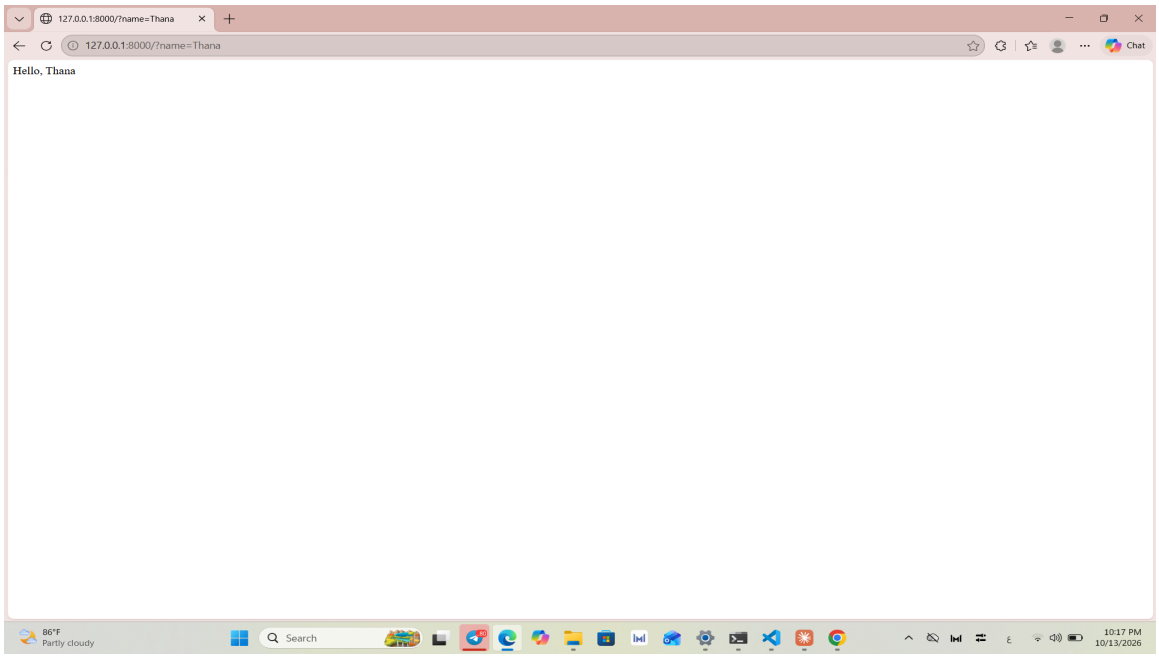


Figure 2: http://127.0.0.1:8000?name=Thana → 'Hello, Thana' (GET parameter captured and rendered)

Task 3 — Second view with URL path parameters

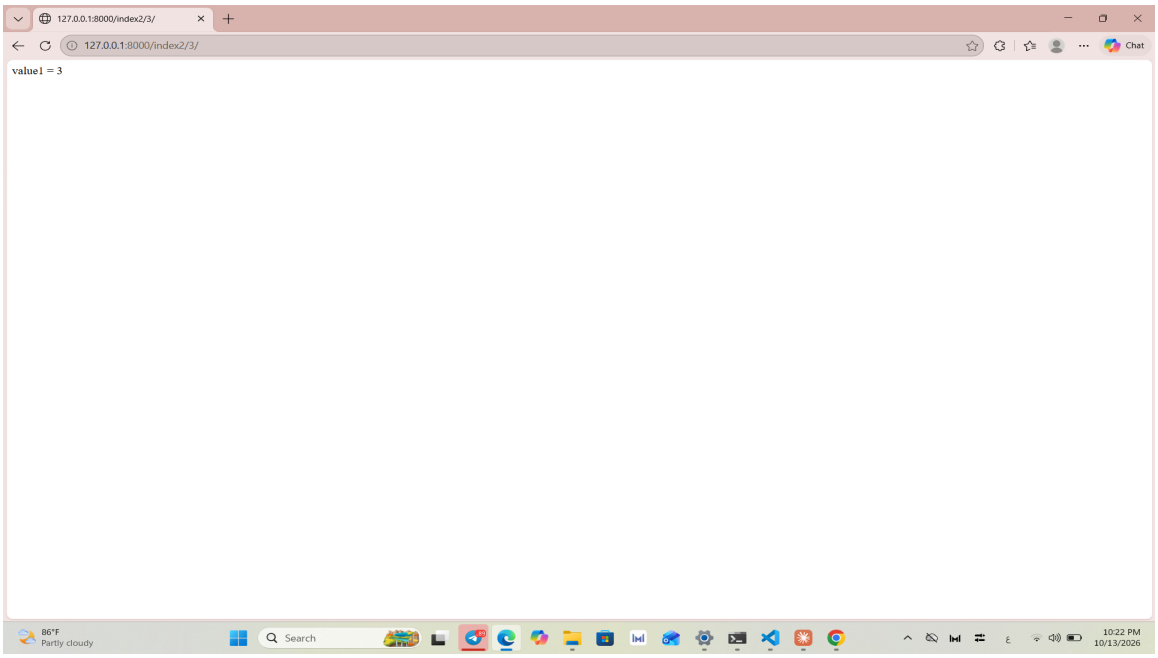


Figure 3: http://127.0.0.1:8000/index2/3/ → 'value1 = 3'

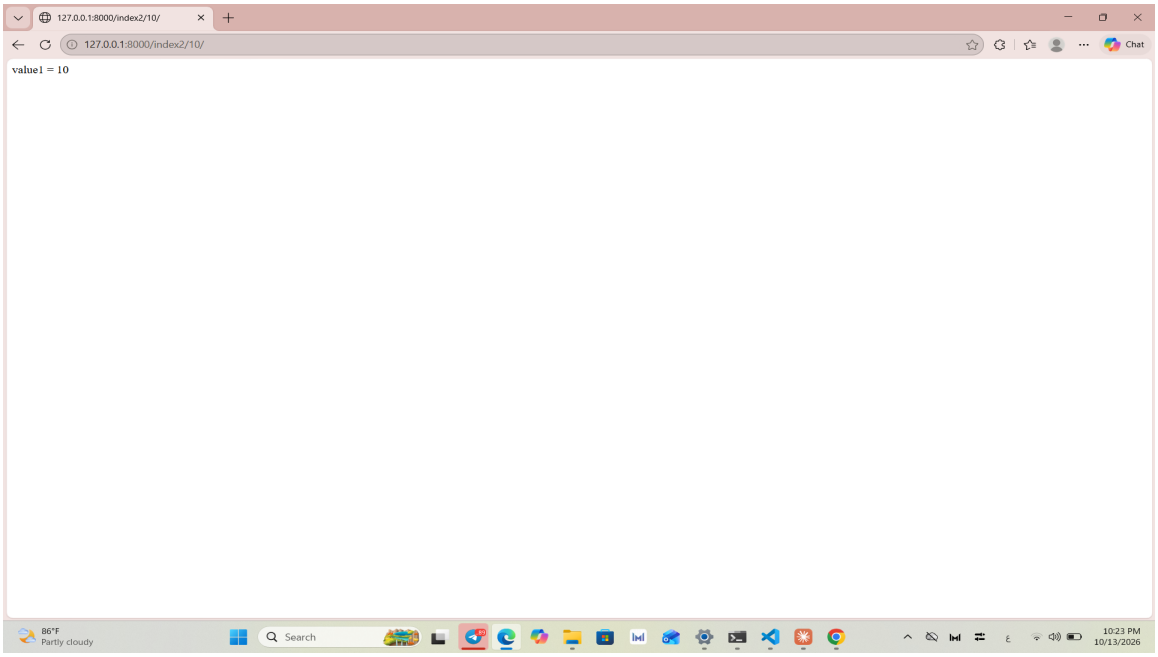


Figure 4: http://127.0.0.1:8000/index2/10/ → 'value1 = 10'

Task 4 & 5 — HTML template with rendered context

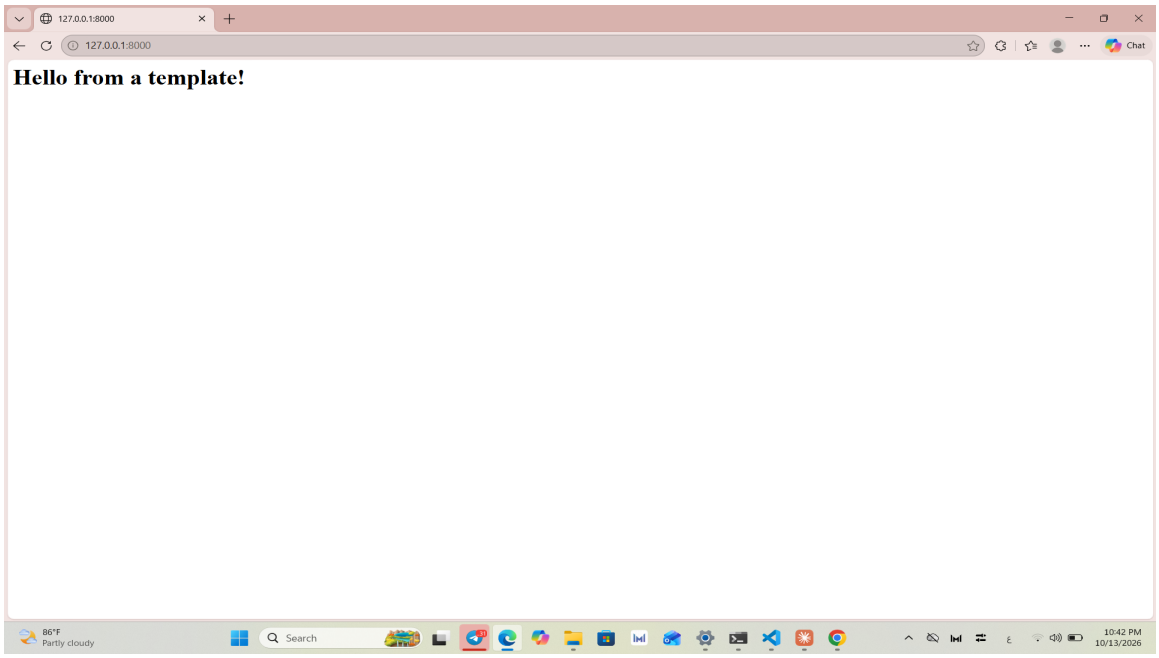


Figure 5: index.html template rendering 'Hello from a template!' via render() and the {{ name }} context variable

Task 6 — Per-app URL configuration (books/, users/)

apps/bookmodule/urls.py and apps/usermodule/urls.py were created and included in the project-level libraryproject/urls.py using Django's include(). All book routes now live under the /books/ prefix, confirmed working in Figures 6-11 below (list, about, links, formatting, listing, tables all resolve correctly under /books/...).

Task 7 — Dynamic book detail view

A viewbook(request, bookId) function was added that looks up a book by ID from an in-memory dictionary and renders bookmodule/show.html with the matching book's data as context.

URL	Rendered Output
/books/123	ID: 123 | Title: Continuous Delivery | Author: J. Humble and D. Farley
/books/456	ID: 456 | Title: Secrets of Reverse Engineering | Author: E. Eilam

Both book detail pages rendered the correct ID, title, and author via the Django template context, confirming Task 7 is complete.

Lab 4 — HTML, CSS & Django Template Tags

Four HTML pages (Home, About, List Books, One Book) were built with inline, internal and external CSS, then integrated into the Django project as static files and templates using {% static %}, {% url %}, {% extends %}, {% block %} and {% include %} template tags. A shared layout (base.html) with a common header and footer was created so every page inherits the same navigation bar and styling, with the active-page highlighted via styles.css.

Home page — layout, navigation bar, inline & external CSS

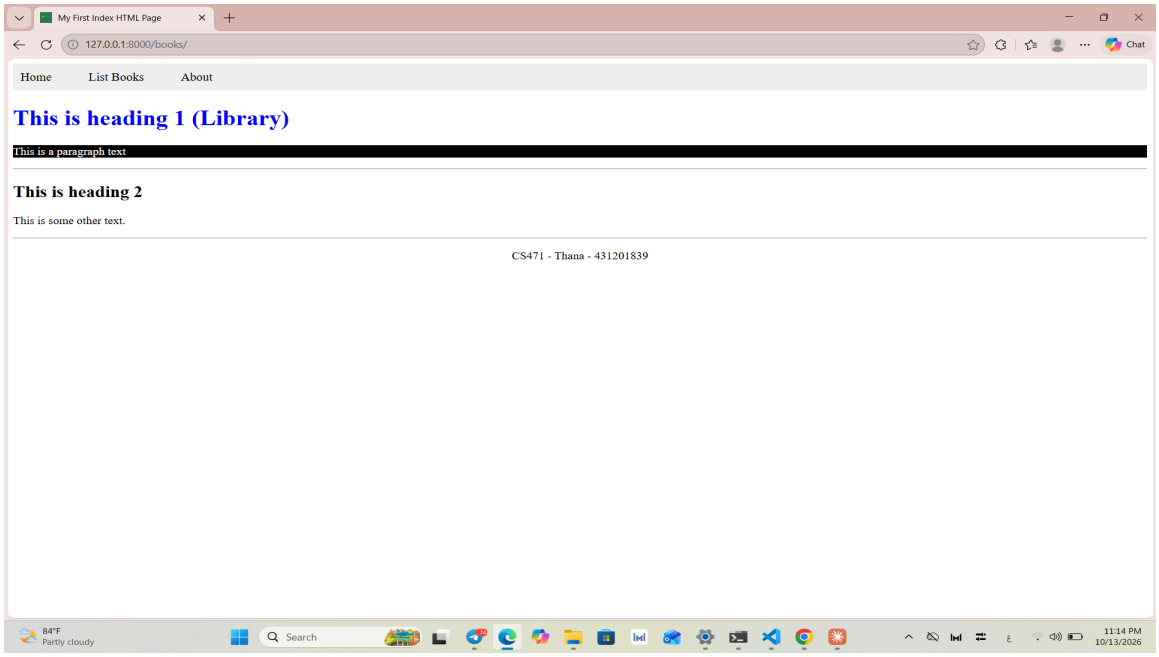


Figure 6: /books/ — shared header/footer layout, inline CSS on heading and paragraph, navigation bar styled from styles.css

List Books page — static images served via {% static %}

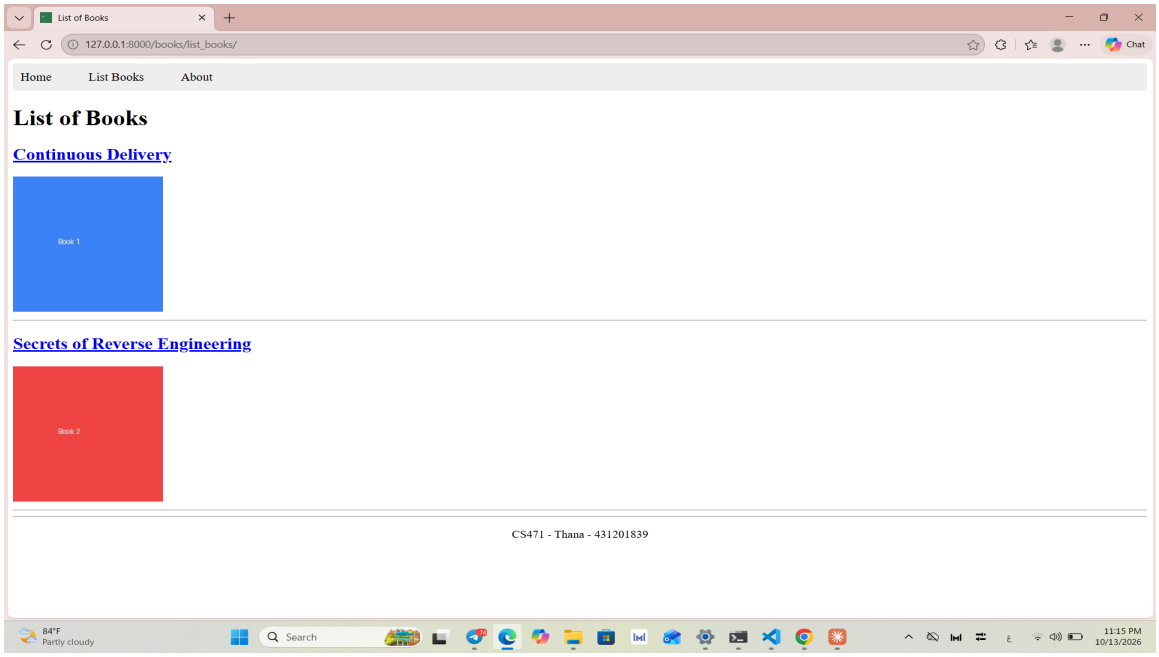


Figure 7: /books/list_books/ — book cover images and links loaded from the apps/static/ folder, each linking to its detail page

Static files configuration

settings.py was updated with STATIC_URL and STATICFILES_DIRS pointing to apps/static, and the CSS file plus four images (qu-icon.png, book1.jpg, book2.jpg, book3.jpg) were placed inside that folder, confirmed working in the figures above (icon favicon, styled nav bar, and rendered book cover images).

Lab 5 — HTML & CSS Fundamentals (Part 2)

Four additional pages were added under the bookmodule app, each demonstrating a different HTML/CSS concept and all inheriting the shared base.html layout built in Lab 4.

Task 1 — Links & link-state styling (/books/html5/links)

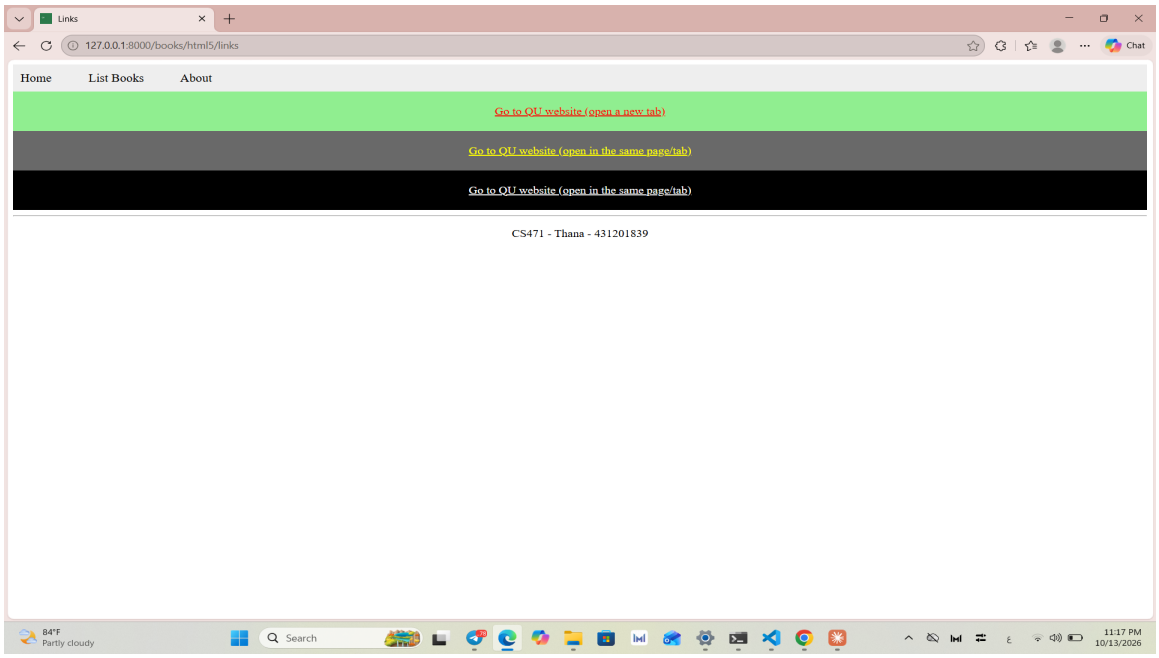


Figure 8: Three styled links (padding, centered text, block display, distinct background/text colors) with visited/hover/active states defined

Task 2 — Nested text-formatting tags (/books/html5/text/formatting)

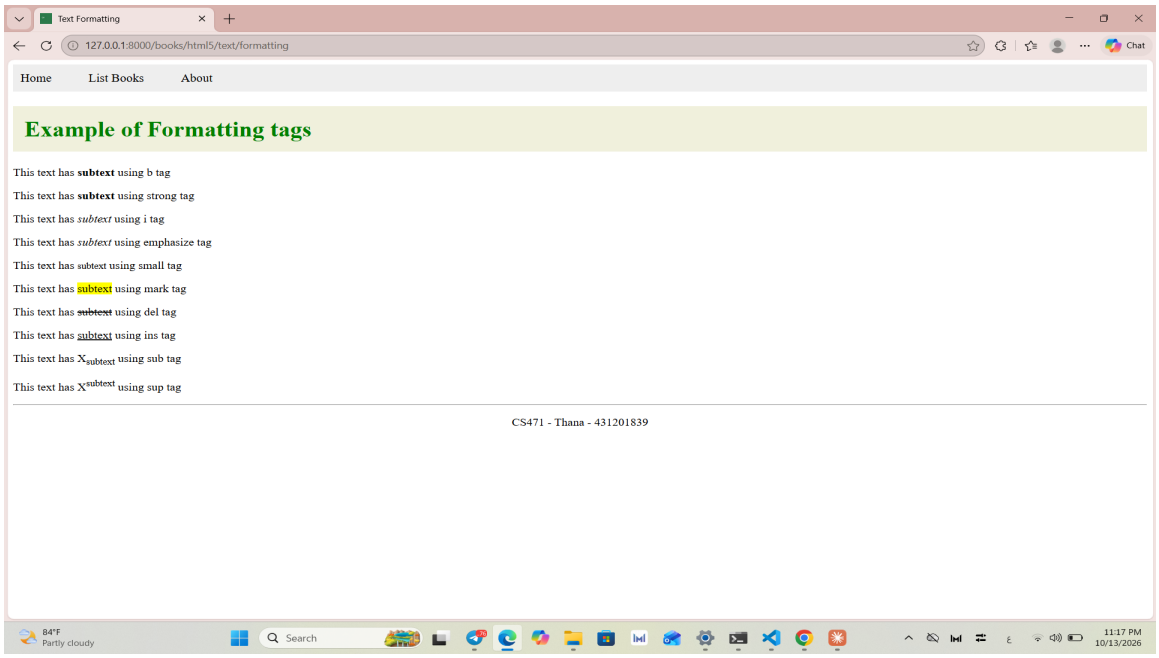


Figure 9: Demonstration of b, strong, i, em, small, mark, del, ins, sub and sup tags, with a styled heading via main CSS

Task 3 — Nested lists (/books/html5/listing)

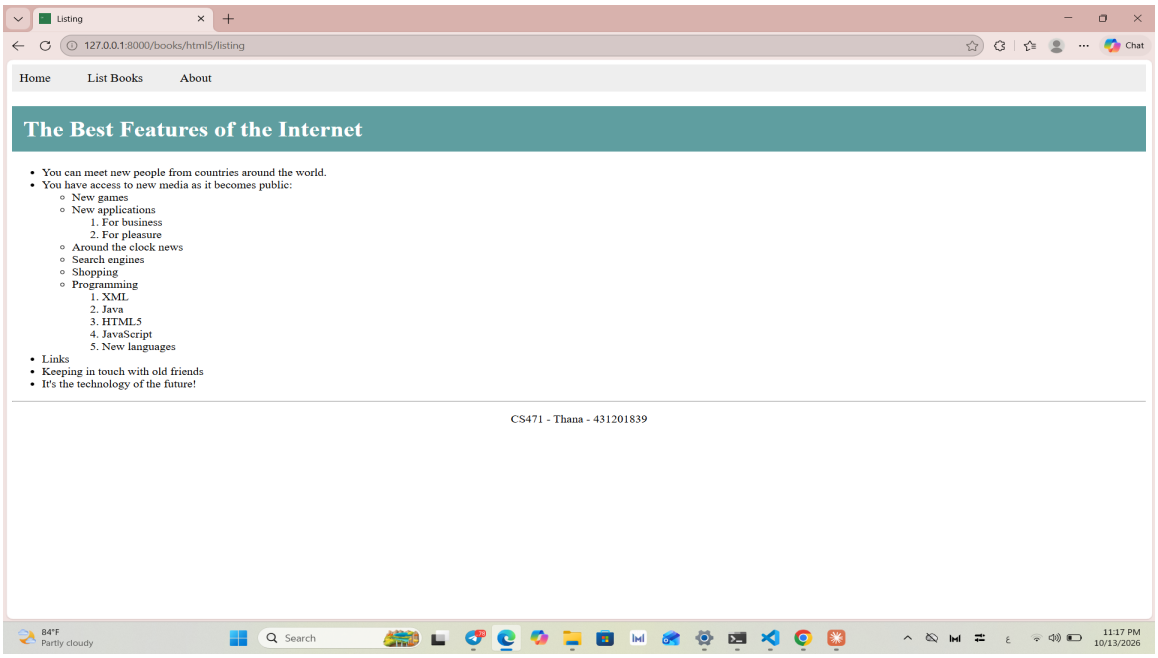


Figure 10: Nested unordered and ordered lists (multi-level) matching 'The Best Features of the Internet' structure

Task 4 — Tables & layout (/books/html5/tables)

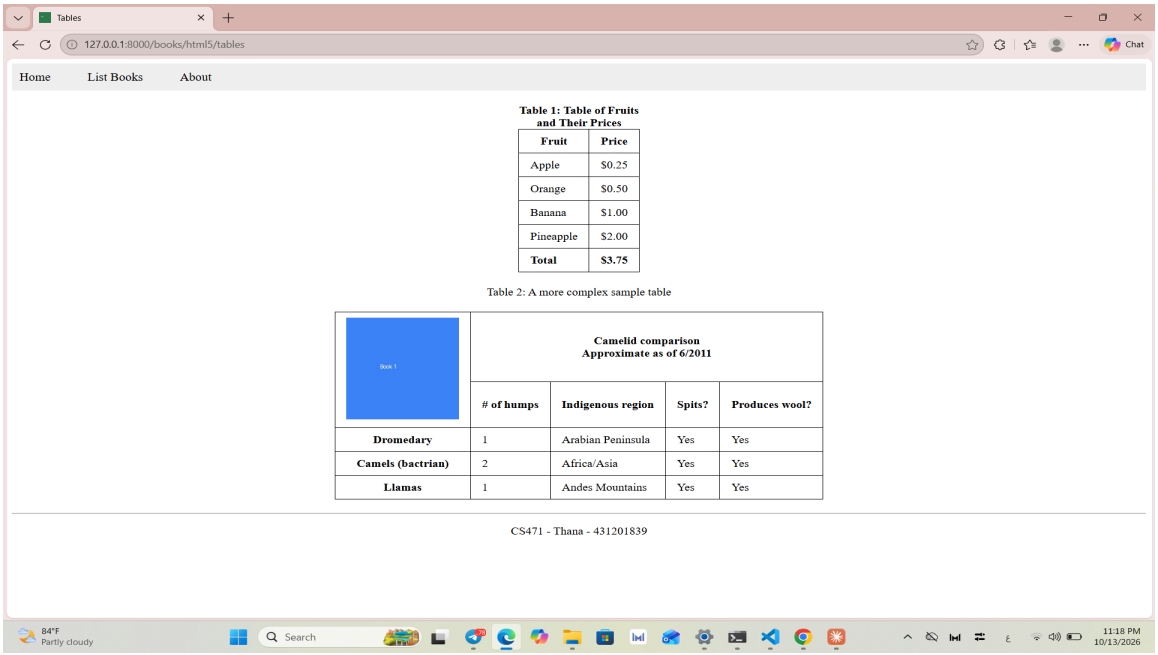


Figure 11: A simple fruit-price table with a totals row, and a complex table using rowspan/colspan with an embedded image

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All source code (views.py, urls.py, templates, static files) for Labs 3, 4 and 5 is available in the GitHub repository below.
https://github.com/Thana111/CS471-Labs